The Aircraft Departure Scheduling Based on Particle Swarm Optimization Combined with Simulated Annealing Algorithm

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Abstract—**Particle swarm optimization combined with simulated annealing algorithm (PSOCSA) was an improved particle swarm optimization algorithm which introduced the simulated annealing (SA) strategy in particle swarm optimization (PSO). It was proposed to solve a mathematical model which is built for aircraft departure sequencing problem in this paper. The correlative implementation techniques and detailed design process of the algorithm were presented. Then the simulation is performed to solve a representative problem using PSOCSA, PSO, and SA. The comparison showed that the PSOCSA algorithm was rational and feasible and more easily converge to the global optimal solution of aircraft departure sequencing problem. Method described in this paper will curtail the consumption of aircraft departure effectively, so it is worth researching it further in the field of airport operations and air traffic control.**

I. Introduction

As a part of air traffic flow management (ATFM), the sequencing problem of aircraft departure and entry attracts people's attention more and more. Especially, with the rapid development of air traffic, the increasing demand of air travel has made the airlines purchase more aircrafts. Under these circumstances, large amounts of congestion are incurred at major airports. Serious congestions not only cause air ways momentous loss and controllers' heavy burden, but also cause air traffic fatal danger sometimes [1-3]. Having a look to other fields where scheduling problems are routinely dealt with, but the constraints of aircraft departure scheduling are more complicated and different from others. So, researches on this field make more sense. So, research about this field has become very urgent. The algorithms for aircraft entry scheduling which contains the aircraft entry sequencing and so on is more mature. Compared with the entry scheduling, there was less research on algorithms related to departure scheduling. So, it is worthy of discussing the departure scheduling problem. In [5], the improved adaptive probabilities of crossover and mutation were introduced in the genetic algorithm to solve departure sequencing problem, but the operation of election, crossing and variation of genetic algorithm was complicated and was troublesome for implementation. Based on the correlative theories and the characteristics of the problem itself, PSOCSA algorithm was designed to solve the aircraft departure sequencing problem in this paper. It was not only efficient and simple, but also easily converged to the global optimal solution. The organization is as follows: Section Ⅱ described the mathematical model of the aircraft departure sequencing problem. Section Ⅲ introduced the particle swarm optimization combined with simulated annealing algorithm. Ⅳ was the algorithm design. Section V was the simulation conditions and simulation results of the PSOCSA algorithm and compared with PSO algorithm and SA algorithm. The conclusion is in the end.

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II. The Mathematical Model of The Aircraft Departure Sequencing Problem

A. The description of Departure scheduling problem

Aircrafts departure refers to the whole process before aircrafts move to the enroute phase of flight from the air-port surface, which consists of flying-off, flying away from the airport and leaving from the terminal area. Generally to say, departure operations include ground operations and air-borne operations. The former, which is mainly discussed in this paper, is a significant source of delay. The ground operations will occur during the departure aircrafts move from the ramp to the runway.

The aircraft departure scheduling is virtually an optimal sequencing of departure aircrafts in a certain period in order to make the Actual capacity of the runway maximal and effectively reduce the time that aircrafts strand in the air over airport. When an aircraft is pulled to the edge of the ramp, it needs to be assigned a taxiway, a runway, and a departure time. Each departure aircraft is usually assigned a runway in advance based on the traffic statistics.

Here, the airport researched is supposed to have two taxiways and one runway. The related departure aircrafts are arranged into many queues and one queue possesses one taxiway, as show in Fig 1. For different airports, the essentials of departure operations are similar in rough, so the research result of this paper is universally applicable to some extent.

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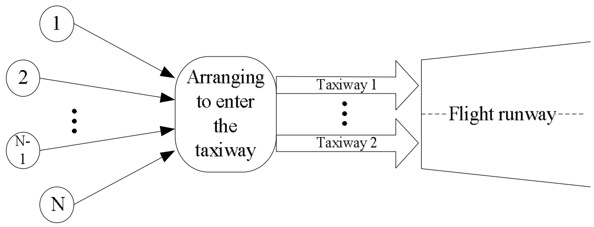

Fig 1.departure of aircrafts

As mentioned above, in order to make the total problem tractable, we consider it as two successive and separate optimization problems.

The first problem is to assign an aircraft to a departure queue. W hen the aircraft is pulled to the ramp, decisions need to be made about which taxiway the aircraft should follow. Usually the assignment criterion is chosen based on the statistics and the related rules. Thus, this problem is often solved artificially and will not be discussed here.

The second problem to be addressed here is the optimal queuing when departure aircrafts entering the flight runway. When the aircrafts reach the pad at the end of the taxiway, they would join one departure queue to enter the runway at last. So the key of this problem is how to select an aircraft to fly first from the heads of the taxiways to minimize the total time costs. The objective is to minimize the total time needed for all related aircraft.

B. Departure sequencing model

To minimize the total time costs, or to maximize the runway throughput, we build a mathematical model[5] for solving the departure scheduling problem as follows:

$$C = \sum_{i=1}^{N} (d_i - d_{i-1}) = d_N - d_0 \qquad (1)$$

where C represents the total time cost, d_i is the departure time of the *ith* departure aircraft, d_0 denotes the departure time of the aircraft that is currently occupying the runway, and d_N denotes the departure time of the last aircraft in the queue composed of N departure aircrafts.

Based on the principle of hydrodynamics, some kinds of separation are supposed to exist between two conterminous aircrafts for safety. Thus, equation (1) is subject to the following constrains:

$$d_i - d_{i-1} \geq \max(m_{i,i-1}, w_{i,i-1}) \quad i = 1, \cdots, N \qquad (2)$$

where $m_{i,i-1}$ indicates the miles-in-trail separation between the *ith* and *(i-1)*th departure aircrafts, and $w_{i,i-1}$ is the wake vortex separation between the *ith* and *(i-1)th* departure aircrafts. Equation (2) occurs because successive departing aircrafts must maintain a minimum separation that is greater than or equal to the maximum among the miles-in-trail separations and the wake vortex separations.

In order to minimize the total time costs, all departure aircrafts should progress one by one and depart without intermission, which means that successive departing aircrafts maintain a minimum separation that is equal to the maximum among the miles-in-trail separations and the wake vortex separations. Then, let $\delta_{i,i+1} = \max(m_{i,i+1}, w_{i,i+1})$, and (1) can be transformed into

$$C = \sum_{i=1}^{N-1} \delta_{i,i+1} \qquad (3)$$

Once the aircrafts are queued on one taxiway, the last departure queue of them is unaltered in most cases according to the "first-come first-serve" principle. That is to say, the aircrafts within one queue must depart in the order in which they are lined up. This constraint is expressed as

$$d_j^k - d_{j-1}^k \geq 0 \quad j = 1, \cdots, N_k ; \quad k = 1, \cdots, K \qquad (4)$$

where d_j^k indicates the departure time of the *jth* aircraft in the *kth* queue, N denotes the number of all departure aircrafts (assume that there are k queues of departure aircrafts, and there are N_k aircrafts in queue k) . Obviously the following equation holds:

$$\sum_{k=1}^{K} N_k = N \qquad (5)$$

If all aircrafts can be reordered at the runway intersections, there would be *N*! possible queues for total aircrafts. However, the number of queues is reduced by (4). This is presented by the following theorem[5].

Theorem 1 Given *k* queues of aircrafts and N_k aircrafts in the *kth* queue, there are $N! \Big/ \prod_{k=1}^{K} (N_k!)$ permissible departure queues.

III. THE PARTICLE SWARM OPTIMIZATION COMBINED WITH SIMULATED ANNEALING ALGORITHM

Introducing the thinking of "simulated annealing" in PSO algorithm which can improve the Local search capability of PSO, PSOCSA algorithm makes particles will not jump out a nearby neighborhood of the optimal solution at the last steps when evaluation values of particles are relatively near. This can accelerate convergence of PSOCSA algorithm.

A. Normal particle swarm optimization algorithm

PSO algorithm[6] began with the original intent to graphically simulate the graceful but unpredictable choreography of a bird flock. By observing the social behavior of animals, people discovered that social sharing of information among population offers an evolutionary advantage, which was fundamental to the form of original PSO. After that, the normal PSO which adds an inertia weight to the speed update equation of the original PSO was proposed by Eberhart and Shi[7]. It solves a problem as follows: first, the particles are initialized randomly. Then, it finds the best solution by iteration. In every step, each particle updates itself by following the two best values, one of which is the best solution found by the particle itself, we call it personal best value, namely *pBest,* and the other is the best solution found by the whole swarm, we call it global best value, namely *gBest.*

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When found the two best values which have mentioned above, each particle updates its velocity v and location x according to the formula as follows:

$$v_{id}^{k+1} = wv_{id}^{k} + c_1 r_1 \left(pbest_{id}^{k} - x_{id}^{k}\right) + c_2 r_2 \left(gbest_{id}^{k} - x_{id}^{k}\right) \quad (6)$$

$$x_{id}^{k+1} = x_{id}^{k} + v_{id}^{k+1} \quad (7)$$

where w is the inertia weight. r_1 and r_2 represent uniform random numbers between 0 and 1. c_1 and c_2 are the cognitive and social scaling parameters which usually between 0 and 2.

B. Simulated annealing algorithm

Simulated annealing (SA)[8] algorithm was proposed firstly by Metropolis and others in early 1980s. It simulates the natural phenomena of annealing of solids. Heating up a solid and allowing it to cold down slowly so that thermal equilibrium is maintained to accomplish annealing of solids. In other words, this process consists of two steps. One is to increase the temperature of the heating bath to a maximum value at which the solid melt. The other is to decrease the temperature of the heating bath gradually until the particles arranging themselves in the ground state of the solid. This ensures the atoms possessing a minimum energy state finally. SA algorithm tries to mimic this process and therefore gets its name. According to the criterion of Metropolis, the probability P of the particles going to the equilibrium state at a temperature point T is computed by the formula: $P = e^{-\Delta E / kT}$ Where, E is the inner energy of the solid at point T , ΔE is the change of E , k is the Boltzmann constant.

Simulating E as value of the target function f and T as control parameter t , SA algorithm is gotten to solve the combination optimization problems: First of all, start with initial solution x_0 and initial value T of control parameter. Next, iteratively repeat the follow steps: 1-Produce new solution, 2-Compute error of object function, 3-Determine whether it can be accepted, 4-Choose to receive or drop. Finally, reduce t gradually until the algorithm stop, and then the current solution is the optimal solution. The anneal process is controlled by cooling schedule which included T , attenuation factor Δt , iterative times L which is the length of Mapkob link at each t and termination conditions S of the algorithm. SA algorithm can be divided into three parts: solution space, object function and initial value.

C. Particle swarm optimization combined with simulated annealing algorithm (PSOCSA)

In the implement of PSOCSA algorithm, the particle's current location $x_i(t)$ will be replaced by its next location $x_i(t+1)$ according to the probability P , which can avoid particles plunge in the Local optimization. Simultaneously, P is decided by the temperature T which decreases gradually with the implementation of the algorithm. That is to say, when the function evaluation at $x_i(t+1)$ is worse than the function evaluation at $x_i(t)$, the probability of replacing $x_i(t)$ with $x_i(t+1)$ will decrease gradually, which will make particles don't jump out the prospective search spaces. It should be illuminated here that the algorithm can get idea searches only the temperature declines sufficient slow. Otherwise, if the temperature declines too fast, the probability P will decrease rapidly accordingly, which will cause particles stagnant in a certain region.

From a visual point of view, if we think normal PSO as a simple mimic of group foraging behavior such as group of birds, then "birds" are cleverer in PSOCSA than "birds" in normal PSO because they "test" with a probability before taking actions rather than go to next location directly.

IV. ALGORITHM DESIGN FOR THE PROBLEM

An optimal sequencing algorithm will be designed based on the model, character and theorem mentioned above to solve the departure problem. This queuing problem belongs to the combinatorial optimization problems. If the enumeration algorithm is used to solve the problem, the computing-load will increase rapidly when the number of departure aircrafts becomes larger which will make the algorithm inefficient even useless. The heuristic approach can also be used to solve this problem. Few of heuristic search algorithms have been proposed for solving this problem in [4]. But they can not guarantee the global optimality of the solution.

Based on the foregoing analysis and Theorem 1, PSOCSA algorithm is designed in this part to solve this queuing problem. Simply, two taxiways and one runway are involved in the study. Two queues on taxiways are called "DQ" and the one departure queue formed eventually is called "GQ". The algorithm design here aims to minimize the total time costs.

1) Code design for particles

In the algorithm, one particle represents a kind of GQ sequence. Because the new GQ sequence, namely the new location of a particle obtained by updating, must maintains the original relative order of the two DQs and have practical use, the common coding methods of PSO can not be used here. A simple and effective coding method is designed based on the characteristic of the algorithm in this paper. All aircrafts in the DQ1 will be marked with"1" and all aircrafts in DQ2 will be marked with "2". By doing this, a particle will be a string composed of "1" and "2" and there are M "1"s and N "2"s. That is to say, the dimension of a particle is $M+N$. Particles will search in a special solution space which consists of many different strings which is made up of "1" and "2" and the length is $M+N$. Here, the first "1" to the *Mth* "1" denote the aircrafts $A_1, A_2, \cdots A_M$ of DQ1, the first "2" to the *Nth* "2" denote the aircrafts $B_1, B_2, \cdots B_N$ of DQ2.

For example, given two DQ, one has 5 aircrafts and the other has 7, that is to say there are 12 aircrafts in all. Suppose there is a traversing path of GQ:

$G_x = (A_1\ B_1\ B_2\ A_2\ B_3\ A_3\ B_4\ B_5\ A_4\ B_6\ A_5\ B_7)$, then the code string of this path is $G_y = (1\ 2\ 2\ 1\ 2\ 1\ 2\ 2\ 1\ 2\ 1\ 2)$.

Apparently, this coding method can guarantee that the relative order of DQ is constant. So, the key to the problem is

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how to restrict the number of "1" and "2" in the code string of GQ.

2) The design of particles' updating operation

Here, particles' updating operations must assure that the number of "1" and "2" in a string namely a particle, is invariant after updating so that the aircrafts in GQ can maintain their original relative order of DQ. The evolution equations (6) and (7) of PSO can not guarantee this obviously, so some constrain should be added.

First, the new location string $x_{id}^{k+1}(d=1,2\ldots,M+N)$ which is obtained by (7) should be round to nearest integer, and then set $x_{id}^{k+1}=\begin{cases}1, if\ x_{id}^{k+1}<1\\ 2, if\ x_{id}^{k+1}>2\end{cases}$.

Next, because the numbers of "1"and "2"in the string have changed, they should be reset again. Suppose that there are a "1"s and b "2"s in the string now, if $a>M$, $c1=a-M$ then find "1" in the string and replace $c1$ of them with "2" randomly; otherwise, if $b>N$, $c2=b-N$, then find "2" in the string and replace $c2$ of them with "1" randomly.

After these operations added, the numbers of "1" and "2" in an updated particle will maintain changeless.

3) Introducing of "simulated annealing"

The highlight of the algorithm is that it introduces the strategy of simulated annealing in PSO. Its main idea is as follows:

First, set a temporary variable marked x_{temp_d}, suppose x_{id}^{k} is a particle of the current population.

$x_{temp\_d}=x_{id}^{k}$

Find one "1" in x_{temp_d} randomly and change it to "2", at same time, find one "2" in x_{temp_d} randomly and change it to "1" too.

While $t>T_{end}$ (t is the current temperature which equal to T_0 at the beginning)

If $f(x_{temp-d})\le f(x_{id}^{k})$ then

$x_{id}^{k}=x_{temp-d}$

$t=t*\beta$ (β is the fading factor which usually is a constant slightly less than 1.00)

Else

$x_{id}^{k}=x_{temp-d}$ according to a probability: P_T

End

End

$v_{kd}^{k+1}=wv_{id}+c_1r_1\left(p_{id}-x_{id}^{k}\right)+c_2r_2\left(g_{id}-x_{id}^{k}\right)$

$x_{id}^{k+1}=x_{id}^{k}+v_{id}^{k+1}$

Based on the analysis above, all steps of PSOCSA are proposed as follows:

STEP 1 Initialization. Set the lengths of each DQ and produce queues of the departure aircrafts randomly, set the maximal iteration time *MaxIter*, the size of the population *POPSIZE*, the initial T_0 and the ending temperature T_{end}, the termination conditions and so on. The current iteration time is *iter* which equal to 0 at the beginning and the initial values of each particle are $X_i^{\,0}=[x_{i,1}^0,x_{i,2}^0,\cdots,x_{i,M+N}^0]\quad \forall i=1,2,\cdots,POPSIZE$ which is produced randomly to form the initial population $P(0)$ and meets

$$\sum_{j=1}^{M+N} j=\begin{cases}M & x_{i,j}^0=1\\ N & x_{i,j}^0=2\end{cases}\qquad \forall i=1,2,\cdots,POPSIZE\,;$$

The initial velocities of each particle are random numbers between -1 and 1.

STEP 2 Compute the personal fitness value ($F_i\quad \forall i=1,2,\cdots POPSIZE$), update the personal best value and global best value and set $iter=iter+1$;

STEP 3 Update the current locations of particles according to the strategies 2), 3) and (6), (7) mentioned above;

STEP 4 Checking the ending conditions ($iter=MaxIter$). If it meets the ending conditions, exit and the current global best value is just the global optimal solution; otherwise, turn to step 2.

STEP 5 Stop the computation and output the correlative values of the optimal individual Such as *gBest* and *f(gBest)*.

V. Simulation Analyses

A. Simulation condition

Four types of aircrafts are designed manually in our simulation, i.e., heavy, large, medium, and small aircrafts, which are denoted by H, L, M, and S, respectively. All the aircrafts depart from one runway in the case of the relative orders of the two DQs are invariable.

TABLE1
WAKE VORTEX SEPARATIONS IN THE SIMULATION MIN

Lead aircraft	Trail aircraft			
	H	L	M	S
H	1.2	2.0	3.0	5.0
L	1.0	1.0	1.2	1.5
M	0.8	0.8	1.0	1.2
S	0.5	0.5	0.8	1.0

Table1 shows the wake vortex separations of the aircrafts. The miles-in-trail separation is not considered here for simple, because adding a constant to the model does not change the nature of the solution.

The algorithm is programmed in MATLAB 7.0 and runs on a PC with 1.86 GHz CPU. $POPSIZE=10$, $MaxIter=100$, $T_0=10$ and $\beta=0.96$.

The departure scheduling problem usually involves a limited time. Especially, here the study refers to two taxiways and one runway, so the aircrafts arranged should not more. Generally, the upper limit of the number of each queue is 20. But the number of aircrafts is not restricted in the simulation of this paper and more aircrafts can be dealt with in fact.

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The research involves different types of departure queues. A representative case is shown in Table 2.

TABLE 2
TWO PRIMITIVE QUEUES OF DEPARTURE AIRCRAFTS

Queue	1	2	3	4	5	6	7	8	9
D1	L	H	S	S	H	H	S	M	H
D2	S	M	H	L	S	M	S	L	L

TABLE 2 CONTINUED

Queue	10	11	12	13	14	15	16	17	18
D1	S	S	H	M	S	S	H	H	
D2	M	L	H	M	S	H	H	H	S

Note: D1 and D2 refer to the first and second queue of departure aircrafts, respectively. Then, D2 (i) means the *ith* aircraft in the second queue.

B. Simulation results

TABLE 3
DEPARTURE QUEUES AND CORRESPONDING TIME TABLES

Random sample		Optimization result 1			Optimization result 2		
Type	Time /min	Type	Queue	Time /min	Type	Queue	Time /min
L	0	S	D2(1)	0	S	D2(1)	0
S	1.5	L	D1(1)	0.5	L	D1(1)	0.5
M	2.3	M	D2(2)	1.7	M	D2(2)	1.7
H	3.1	H	D1(2)	2.5	H	D1(2)	2.5
H	4.3	H	D2(3)	3.7	H	D2(3)	3.7
S	9.3	L	D2(4)	5.7	L	D2(4)	5.7
S	10.3	S	D2(5)	7.2	S	D2(5)	7.2
L	10.8	M	D2(6)	8	M	D2(6)	8
S	12.3	S	D2(7)	9.2	S	D1(3)	9.2
M	13.1	S	D1(3)	10.2	L	D2(7)	10.2
H	13.9	L	D2(8)	10.7	S	D1(4)	11.2
S	18.9	S	D1(4)	12.2	S	D2(8)	11.7
H	19.4	H	D1(5)	12.7	H	D1(5)	12.7
S	24.4	H	D1(6)	13.9	H	D1(6)	13.9
L	24.9	L	D2(9)	15.9	L	D2(9)	15.9
L	25.9	M	D2(10)	17.1	M	D2(10)	17.1
M	27.1	S	D1(7)	18.3	S	D1(7)	18.3
L	27.9	L	D2(11)	18.8	L	D2(11)	18.8
M	29.1	M	D1(8)	20.0	M	D1(8)	20
H	29.9	H	D2(12)	20.8	H	D1(9)	20.8
S	34.9	H	D1(9)	22.0	H	D2(12)	22
H	35.4	M	D2(13)	25.0	M	D2(13)	25
S	40.4	S	D1(10)	26.2	S	D1(10)	26.2
H	40.9	S	D2(14)	27.2	S	D1(11)	27.2
M	43.9	S	D1(11)	28.2	S	D2(11)	28
M	44.9	H	D2(15)	28.7	H	D2(15)	28.7
S	46.1	H	D2(16)	29.9	H	D1(12)	29.9
S	47.1	H	D1(12)	31.1	H	D2(16)	31.1
H	47.6	H	D2(17)	32.3	M	D2(17)	32.3
S	52.6	M	D1(13)	35.3	M	D1(13)	35.3
H	53.1	S	D1(14)	36.5	S	D1(14)	36.5
H	54.3	S	D2(18)	37.5	S	D2(18)	37.5
H	55.5	S	D1(15)	38.5	S	D1(15)	38.5
H	56.7	H	D1(16)	39	H	D1(16)	39
S	61.7	H	D1(17)	40.2	H	D1(17)	40.2

For the two primitive queues shown in Table 2, some optimization results are obtained by using the model and PSOCSA algorithm. A comparison between the optimization results and a random sample is given in Table 3. It is clear that the total departure time is reduced successfully from 61.7 min for random sample to 40.2 min after optimization.

Because only four types of aircraft are involved, so the optimization result is not exclusive naturally. In other words, there will be different optimization departure sequences with same total time costs.

Results in Table 3 shows that the last optimization departure sequence still maintains the original relative orders of the two DQs and the total departure time consumed is minimal. Moreover, there is not long interval between the departing of the last aircrafts of the two queues, namely the departure of the aircrafts in the two queues are well-proportioned in the last optimization departure sequence. What mentioned above indicates the correctness and validity of the model and algorithm described in this paper.

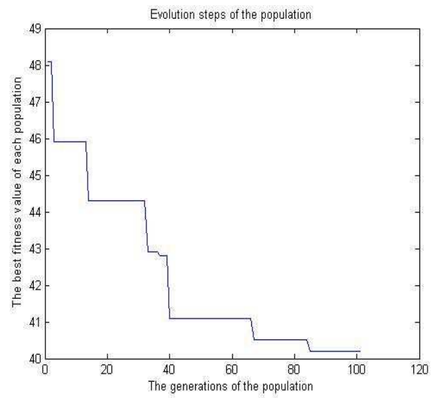

Fig. 2 Curve change of population's best fitness value

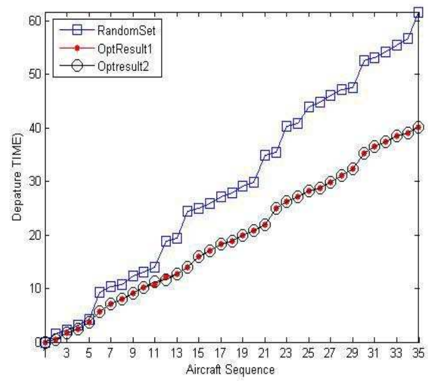

Fig. 3 Chart of the aircrafts departure time before and after optimization

As shown in Fig. 2, the algorithm was able to skip from local optimization (e.g. the value 41 and so on in Fig. 2) and obtain the global optimization finally; Fig. 3 is an explicit expression corresponding to Table 3, which shows: 1) The distribution of the departure time according to the optimization results is better-proportioned than that according to random sample, namely the case of big wake vortex separations would be eliminated by optimization, such as the small aircraft would not be allowed to follow the heavy one. 2) The global optimization does not mean that the last optimization departure sequence is exclusive. For example, swapping the two adjacent aircrafts which are the same types with the relative order of the two DQs unchanging, do not change the optimizing result. 3) The departure times of some aircrafts in the random sample may be earlier than that in the optimization results, which implies that the global optimization does not mean optimization at every step.

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TABLE 4
COMPARING OF TEN TESTS RESULTS OF PSOCSA, PSO, AND SA

Gbest \ time	1	2	3	4	5	6	7	8	9	10	ave
PSO	40.3	41.1	40.3	42.9	43.7	40.3	43.7	41.5	41	40.8	41.56
SA	43.5	40.9	43.7	40.3	43.5	40.2	40.2	42.8	41	40.2	41.63
PSOCSA	40.2	40.4	40.2	40.2	40.2	40.3	40.3	40.2	40.2	40.2	40.24

Table 4 shows that the global best values and the average value of ten tests using PSO algorithm, SA algorithm, and PSOCSA algorithm respectively with the same initial conditions. With comparing, the conclusion can be drawn that the PSOCSA algorithm can converge to the global optimization more steadily and rapidly than PSO and SA.

VI. CONCLUSION

A mathematical model is described for solving the aircraft departure scheduling problem in this paper and PSOCSA algorithm which combines PSO with SA is applied to solve the problem. Then the related implementation techniques are given. Finally, the simulation is performed and the results are compared with PSO and SA to verify the effectiveness of the method.

The method proposed in this paper can effectively decrease the total time costs of the aircraft departure with a well-proportioned departure time schedule. The model and the algorithm are all effective. The algorithm is not only simple for programming, but also can guarantee the global optimization solution. Hence it is very useful for air traffic flow management.

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